

Design Techniques for Wideband Single-Bit Continuous-Time $\Delta\Sigma$ Modulators With FIR Feedback DACs

Pradeep Shettigar, *Member, IEEE*, and Shanthi Pavan, *Senior Member, IEEE*

Abstract—We give design considerations for single-bit continuous-time Delta-Sigma modulators (CTDSMs) with FIR feedback DACs. These modulators have the low jitter sensitivity and high linearity properties characteristic of a multibit modulator, while using a simple one-bit quantizer, thereby combining the advantages of single-bit and multibit operation. We propose a method to compensate the loop for the delay introduced by the FIR-DAC. The efficacy of our architectural and circuit techniques is borne out by measurement results from a modulator that achieves about 71-dB SNDR in a 36-MHz bandwidth while consuming only 15 mW from a 1.2-V supply. Implemented in a 90-nm CMOS process and sampling at 3.6 GS/s, the CTDSM has a figure of merit (FoM) of 72.7 fJ/lv, while occupying 0.12 mm².

Index Terms—Analog-to-digital conversion, continuous-time, oversampling, quantization, sigma-delta.

I. INTRODUCTION

CONTINUOUS-TIME delta-sigma modulators (CTDSMs) have been receiving increasing attention as they are particularly suited for implementation in scaled CMOS technologies. Several high-speed designs achieving signal bandwidths in the tens-of-megahertz range have been reported recently [1]–[6]. When compared with Nyquist rate converters based on switched-capacitor techniques, CTDSMs offer several advantages—their input impedance is resistive, simplifying the design of the ADC driver circuitry. If the DAC is properly designed, they have the implicit anti-aliasing property. The gain requirements of the opamps are relaxed, as the gain needed is “shared” by several amplifiers. Further, since the analog circuitry operates in continuous time, the entire waveform is of relevance, unlike in a switched-capacitor-based design, where only the sampled values are important. Thus, power-efficient opamp topologies (which may not have good settling properties) can be used in CTDSMs.

Most modulators achieving wide bandwidths have used multibit quantizers [1], [2], [6]. With a many-level quantizer, the gain of the quantizer is close to unity (without overload). This means that an aggressive noise transfer function (NTF) can be

chosen. Thus, the over-sampling ratio (OSR) needed to achieve a desired in-band signal-to-quantization noise ratio (SQNR) can be decreased, resulting in relaxed circuit speed requirements. If a nonreturn-to-zero (NRZ) DAC is used, the sensitivity of the converter to clock jitter is reduced. Further, since the loop filter processes the shaped quantization noise (which is reduced by virtue of a small step size), its linearity requirements are greatly relaxed. While the factors described above tend to favor a multibit loop, several issues need to be addressed while realizing such a modulator at high speeds. The key design difficulty lies in the implementation of the quantizer (ADC-DAC cascade). As the number of quantizer bits is increased, the number of comparators increases, but more importantly, the maximum random offset tolerable in each comparator also decreases, necessitating a larger area per comparator and/or complex offset correction schemes. The input capacitance of the ADC, which needs to be driven by the loop filter, also increases, thereby increasing the quantizer delay and power dissipation of the loop filter. Clock distribution also consumes significant power. Overall, the power and area of the ADC increase exponentially. Due to these factors, multibit designs have typically limited the number of quantizer bits to four [1], [6].

The feedback DAC has its set of problems. Dynamic element matching (DEM) techniques, which are commonly used (at low speeds) to counter DAC mismatch, cannot be employed, since the delay introduced by the extra circuitry is unacceptable. Calibration usually results in complex logic. To avoid these problems, several high-speed designs reported recently have used DACs with inherently good linearity, sometimes resorting to the use of a higher supply voltage just to achieve adequate element matching (see [1], [6]).

A belief commonly held by designers is that the first integrator consumes the most power in a CTDSM design. This is generally true at low speeds. At high clock rates, however, the quantizer consumes most of the power due to the factors discussed above. One could argue that the smaller OSR needed in a multibit design should reduce power dissipation as the sampling speed is lower. However, the reduction in the sampling rate is not in proportion to the increased number of comparators. For instance, increasing the number of levels from 8 to 16 in a fourth-order modulator reduces the sampling rate needed to achieve a SQNR of 95 dB by only about 25%. Turning this observation upside down, we see that *reducing the number of quantizer levels*, and making up for the loss in SQNR by increasing the sampling rate, should result in lower power dissipation. An extreme case of this approach is to use a two-level

Manuscript received April 15, 2012; revised July 12, 2012; accepted August 13, 2012. Date of publication November 12, 2012; date of current version December 21, 2012. This paper was approved by Guest Editor Marco Corsi.

The authors are with the Department of Electrical Engineering, Indian Institute of Technology, Madras, Chennai 600036, India.

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/JSSC.2012.2217871

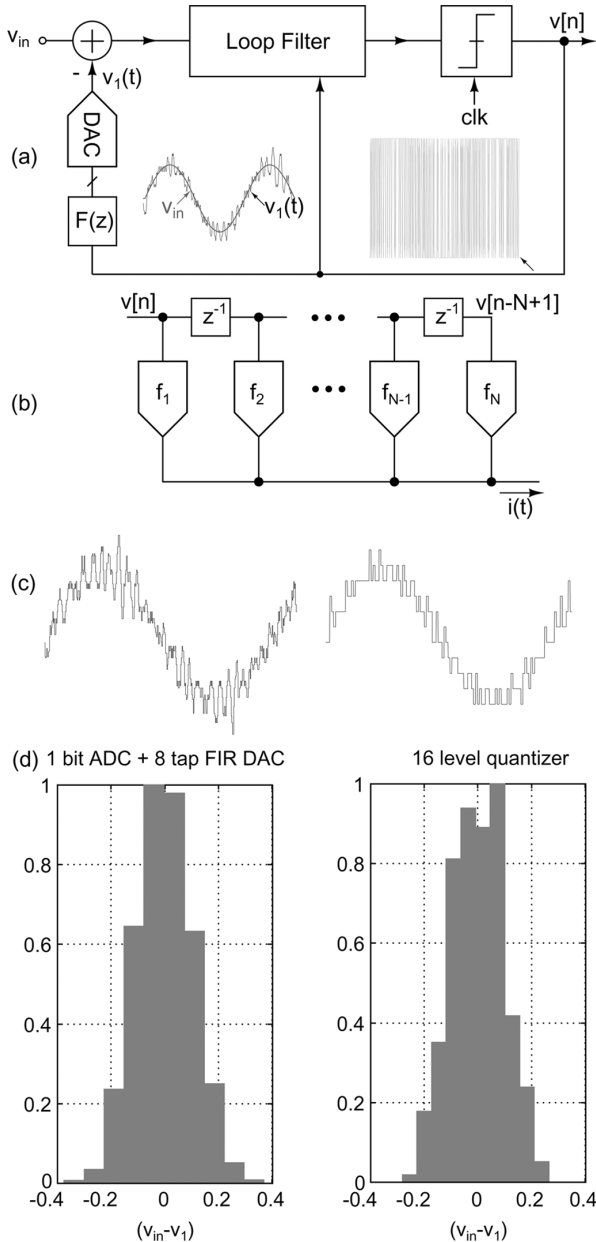


Fig. 1. (a) Single-bit CTDSM with an FIR DAC. (b) FIR DAC implemented using the semi-digital approach. (c) Feedback waveforms in modulators with a 1-b, 8-tap FIR DAC and a 4-b quantizer. The sampling rate of the multibit design is a third of the single-bit design. (d) Histogram of the shaped noise processed by the loop filter. The quantizer output range is assumed to be $[-1, 1]$.

quantizer, where simulations show that a fourth-order modulator with a single-bit quantizer needs to sample at thrice the sampling frequency of a design using a 15-level quantizer (and an OBG of three) to achieve the same peak SQNR of about 95 dB.

A single-bit design has several attractive features—the ADC and clock generator implementations are simple. Since the number of comparators are reduced, the power dissipation of the quantizer is small, though the sampling rate is higher. Due to the inherently linear feedback DAC, DEM is not necessary. This reduces excess loop delay, enabling a higher sampling rate (which is necessary anyway to obtain the desired SQNR). However, several issues must be borne in mind while using such an architecture. The most important concern in a wideband

single-bit modulator is data-dependent jitter due to comparator metastability. Jitter perturbs the width of the feedback DAC pulse (assumed to be of the NRZ kind) and degrades the in-band SNR [7]. Clock jitter also degrades performance through the same mechanism. The severity of these issues can be appreciated when compared with a multibit modulator operating at a lower sampling rate. Apart from the full-scale swing of the feedback signal in the single-bit case (as opposed to a fraction of that in a multibit design), the timing uncertainty in the edges is a larger fraction of the bit period. One way of addressing the jitter problem is to use a switched-capacitor feedback DAC, where the charge injected into the loop filter every cycle remains largely constant irrespective of clock jitter. However, the impulsive currents injected by such DACs can greatly increase the linearity requirements of the loop filter. More importantly, the implicit anti-aliasing property of a CTDSM is lost [8]. Another ramification of single-bit feedback is that the loop filter must process a larger signal, thereby placing increased demands on its linearity and has to be achieved by increasing power dissipation. Finally, the out-of-band gain (OBG) of the NTF must be restricted to about 1.5 for stable operation [9]. This restricts the maximum SNR achievable for a given OSR.

From the discussion above, it is seen that a multibit design is attractive from the viewpoint of reduced data-dependent/clock jitter sensitivity and loop filter linearity while having the disadvantage of needing a power-hungry quantizer. On the other hand, a single-bit design is advantageous in the sense of a simple quantizer, but jitter severely degrades performance, and more power must be dissipated to achieve adequate linearity in the loop filter.

In this work, we attempt to combine the benefits of single bit and multi-bit operation into one architecture by using an FIR DAC as the feedback element. This idea is not new (see [10] and [11] and the references therein), but does not seem to have been employed in high-speed designs. We show that a modulator using a single-bit ADC and an FIR DAC inherits the low jitter sensitivity and high-linearity property of a multibit design and the simplicity of the ADC inherent in a single-bit design—thereby resulting in a power-efficient CTDSM. As a case in point, we present the design and measurement results of a single-bit modulator that achieves about 71-dB SNDR in a 36-MHz bandwidth while consuming only 15 mW from a 1.2-V supply. Implemented in a 90-nm CMOS process and sampling at 3.6 GS/s, the CTDSM has an FoM of 72.7 fJ/1v, while occupying 0.12 mm² [12].

The remainder of this paper, which discusses design issues in FIR DAC-based CTDSMs, is organized as follows. Section II describes the FIR DAC and compares its performance with that of a multibit DAC. In Section III, we discuss the architecture of our modulator and the technique employed to compensate for the excess delay introduced by the FIR DAC. Circuit design details are given in Section IV. Experimental results from our prototype modulator are given in Section V. Section VI concludes the paper.

II. FIR DAC

Fig. 1(a) shows the block diagram of a single-bit CTDSM with an FIR feedback DAC. A single-bit ADC is used, and an FIR filtered version of this sequence drives the feedback DAC.

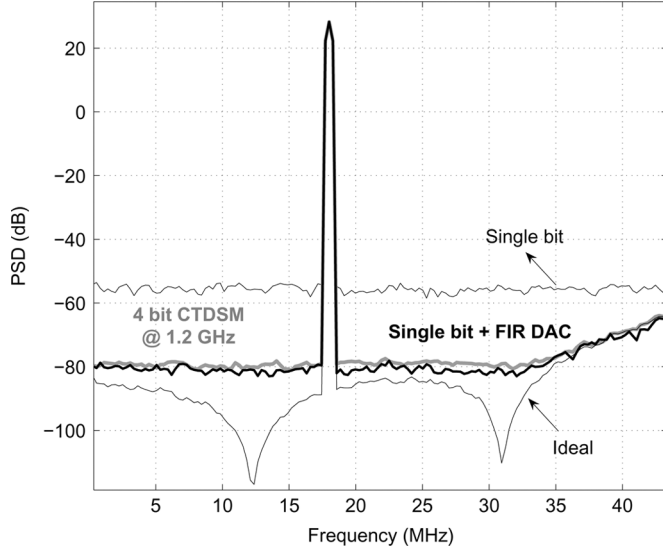


Fig. 2. Effect of white clock jitter (0.3 ps rms) on the in-band spectrum. The CTDSM with the FIR DAC performs marginally better than its multibit counterpart.

This results in a multilevel feedback waveform, as shown in the figure. Notice the similarity of $v_1(t)$ to the feedback waveform in a modulator that employs a multibit quantizer. The transfer function of the FIR filter is denoted by $F(z)$, which introduces delay into the loop. The CTDSM has to be stabilized by accounting for $F(z)$, which is accomplished by feed-ins into the loop filter. Implementing the FIR DAC as diagrammed in Fig. 1(a) has problems with element mismatch in the DAC. Recognizing that $v[n]$ is a two-level sequence, a linear DAC-filter combination can be realized using the semidigital approach [13], as shown in Fig. 1(b). Here, the delays are implemented digitally, while the individual DAC outputs (which are assumed here to be currents) are weighted and summed in the analog domain. DAC mismatch modifies the transfer function of the filter, but does not influence linearity.

We now compare the performance of two CTDSMs designed to achieve the same performance (peak SQNR of about 95 dB) in a 36 MHz bandwidth. One is a single-bit modulator using an 8-tap FIR DAC operating at 3.6 Gs/s, while the other is a multibit design that employs a four bit quantizer, but operates at 1.2 GS/s. The Out of Band Gain of the NTF of the multibit modulator is 3. A full scale of $[-1, 1]$ and NRZ DACs are assumed in both cases. Fig. 1(c) and (d) compare the feedback DAC waveforms and the histogram of the shaped quantization noise respectively. The RMS values of the shaped noise waveforms in both cases virtually the same, indicating that the performance of both modulators with respect to clock jitter and loop filter linearity must be comparable.

Fig. 2 shows the in-band PSDs of the two modulators when the clock has 0.3 ps rms white jitter. The ideal PSD and that of a single-bit CTDSM (without an FIR DAC) are also shown for reference. Thanks to the high frequency noise suppression of the FIR DAC, we see that the noise due to jitter is marginally smaller than that in the multibit quantizer.

Next, we compare the linearity of the multibit and FIR-DAC based CTDSMs when the input integrator in the loop filter is

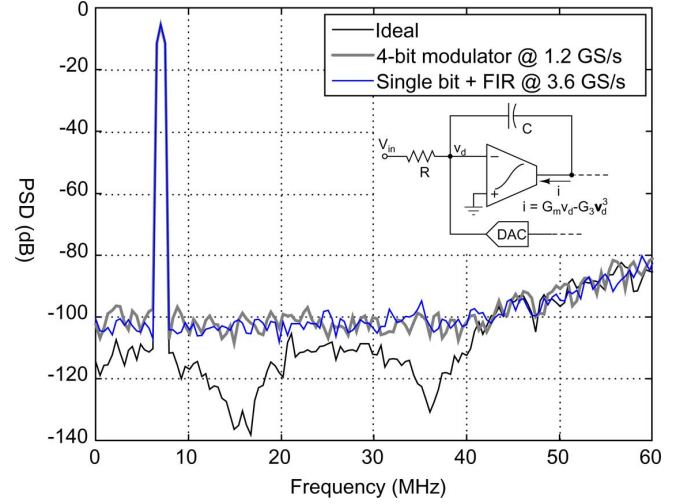


Fig. 3. In-band PSDs of a 4-b CTDSM with $f_s = 1.2$ GS/s and a 1-b FIR-DAC-based CTDSM with $f_s = 3.6$ GS/s. The input integrator is weakly nonlinear, with $R = 1$ k Ω , $G_m R = 20$, and $G_3 = 20$. The PSD for an ideal (linear) modulator is also shown for reference.

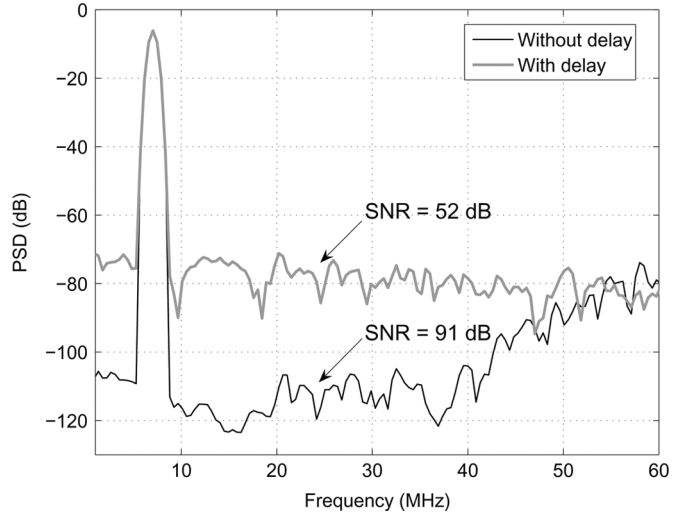


Fig. 4. Comparing the PSDs of a single-bit modulator with and without an explicit delay of one clock cycle.

made weakly nonlinear. The integrator is assumed to be of the active-RC kind, with a single-stage OTA whose output current of the form $i_{out} = G_m v_d - G_3 v_d^3$. It can be shown [14] that weak loop-filter nonlinearity causes the demodulation of out-of-band quantization noise to low frequencies, thereby raising the in-band noise floor. From Fig. 3, we see that the in-band noise floor has increased in both CTDSMs cases by approximately the same amount, indicating that similar linearity is required of the loop filters in both designs to achieve a given performance.

Comparator metastability causes a data-dependent delay in the feedback DAC waveform, thereby influencing the width of the feedback pulse. While this is a problem with multibit and single-bit CTDSMs, it is particularly so in the single-bit case, as described in the introduction. With an FIR DAC, this can be mitigated as follows. Referring to Fig. 1(b), we see that the signal from the first tap of the filter $f_1 v[n]$ is most severely affected by data-dependent jitter. Subsequent flipflops see a much

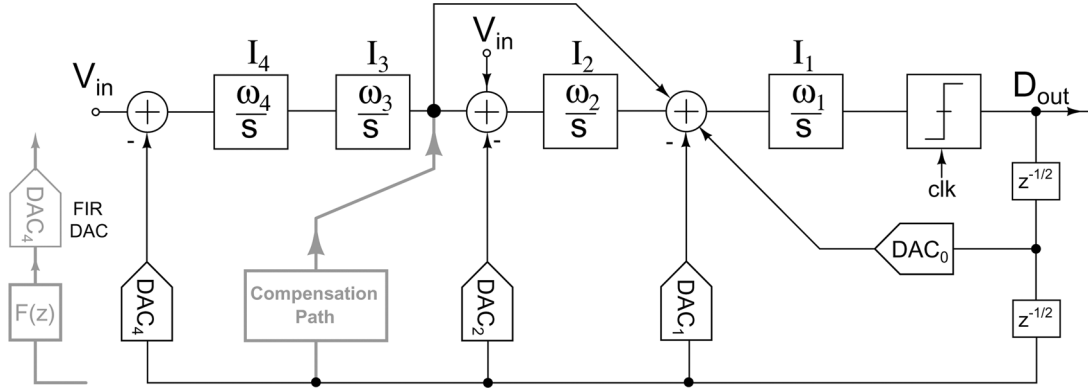


Fig. 5. Modified CIBF structure as the starting point for the architectural development. DAC₄ is replaced by an FIR-DAC, and the loop filter is stabilized for the excess delay by a compensation block shown in gray.

larger input signal, causing the regeneration time to be exponentially reduced. Thus, SNR degradation due to metastability can be avoided by setting the first tap of the FIR filter to zero, which is equivalent to adding an explicit one cycle delay in the FIR path. Fig. 4 compares the PSDs of a modulator with an FIR DAC (sampling at 3.6 GS/s), with and without a delay of one clock cycle. We see that the in-band SNR (in a 36-MHz bandwidth) without delay degrades by about 40 dB due to metastability.

The FIR DAC, therefore, mitigates the drawbacks of a conventional single-bit modulator to a large degree. Due to single-bit operation, the ADC is simple and power-efficient, and no DEM or calibration is needed in the feedback DAC. It is thus seen that an FIR DAC-based single-bit CTDSM combines the merits of multibit and single-bit operation, without the disadvantages of either.

III. MODULATOR ARCHITECTURE

A fourth-order NTF with an OBG of 1.5 and an OSR of 50 results in a peak SQNR of about 95 dB in a 36-MHz bandwidth. The starting point for the architecture used to realize this NTF is a modified cascade of integrators with feedback (CIBF) loop, as shown in Fig. 5. The advantage of a CIBF structure is that the high-speed path around the quantizer (ADC-DAC₀-I₁) is decoupled from the high precision path (ADC-DAC₄-I₄-I₃-I₂-I₁), allowing independent optimization of the components comprising these paths.

For the time being, ignore the parts of the diagram shown in gray. A conventional CIBF modulator has four feedback DACs, one feeding into each integrator. This causes each integrator output to have a component dependent on the modulator input, necessitating large integrating capacitors and reducing the gains of the integrators in the signal band. This is particularly severe in I₄. To prevent this, the $1/s^3$ path of the loop filter is realized by feeding the output of I₃ to the input of I₁. To reduce the output swing of I₃, a fraction of the modulator input is added to the input of I₂.

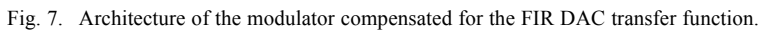
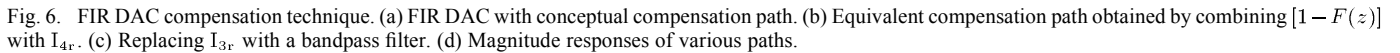
Comparator metastability and data-dependent jitter are problematic in a high-speed design such as this. To address this, the feedback DACs DAC₁, DAC₂, and DAC₄ are driven by the ADC output sequence delayed by a full clock cycle. The extra clock cycle delay would result in an unstable modulator. A common way of solving the excess delay problem is to add

a direct path around the quantizer. However, this necessitates a summing amplifier at the loop filter output, which consumes power and adds delay to the loop. To prevent this, excess delay compensation is realized by a return-to-zero (RZ) DAC (DAC₀) that injects a signal into the input of integrator I₁ [1]. Extra amplifiers are not needed with this approach, since the integrators are of the active-RC variety, where summation of signals at the integrator inputs is accomplished in the current domain. To keep the NTF the same, the coefficient of DAC₁ has to be appropriately adjusted to account for the increased $1/s$ component due to DAC₀.

To reduce the jitter sensitivity of the CTDSM and relax the linearity requirements of I₄, an DAC₄ is replaced by an FIR DAC, whose transfer function is denoted by $F(z)$, as indicated in gray to left of DAC₄ in Fig. 5. Introduction of the FIR filter causes the response of the path DAC₄-I₄-I₃ to change, rendering the modulator unstable. The loop has to be stabilized, and this is done by adding a compensation path in parallel to the $F(z)$ -DAC₄-I₄-I₃ path as explained with the aid of Fig. 6. Fig. 6(a) shows the FIR DAC $F(z)$ driving I₄-I₃. The DAC is excited by the quantizer output, comprising the input signal and shaped quantization noise. The dc gain of $F(z)$ is unity. The transfer function of the cascade $F(z)$ -I₄-I₃ (signal path (1)) falls as $1/f^2$ at low frequencies—but, more significantly, attenuates high frequencies, as shown in Fig. 6(d). In principle, this deficit at high frequency can be made up by using a compensation path consisting of I_{4r} and I_{3r} (which are replicas of I₄ and I₃), driven by an FIR DAC with transfer function $[1 - F(z)]$ (Fig. 6(a), between p and q). The frequency response of the compensating path (2) is shown in gray in Fig. 6(d).

The compensation path introduces several problems. The jitter noise added by this path completely annuls the jitter noise advantage obtained through the use of $F(z)$. Further, dc offsets in I_{4r} and I_{3r} are bound to saturate the modulator. The jitter noise issue of the compensation path can be mitigated by observing the following. Since $[1 - F(z)]$ and I_{4r} have a zero and pole at dc, respectively, they can be combined into one lowpass FIR DAC $E(z)$ as shown in Fig. 6(b). This way the jitter noise injected by $E(z)$ is only amplified by one integrator rather than two. However, the dc offset problem remains.

$E(z)$ is a lowpass transfer function driven by the quantizer output v , which consists of the modulator's input signal V_{in}



An eight-tap FIR implementation was chosen for DAC₄. Rather than use identical weights for all taps, they were optimized assuming that the jitter spectrum of the clock source falls off as $1/f^2$. The transfer function of the (linear phase) FIR filter is as follows:

Calculations along the lines of [7] show that the eight-tap FIR DAC results in a jitter sensitivity similar to that of a four-bit modulator operating at a third of the sampling rate. The input integrator I_4 integrates the difference between the modulator input and the FIR-filtered shaped quantization noise. Thus, when compared with a single-bit modulator without FIR feedback, the linearity required of I_4 is greatly relaxed. Fig. 8 compares the NTF of the compensated modulator with a maximally flat NTF with an OBG of 1.5. The effect of adding

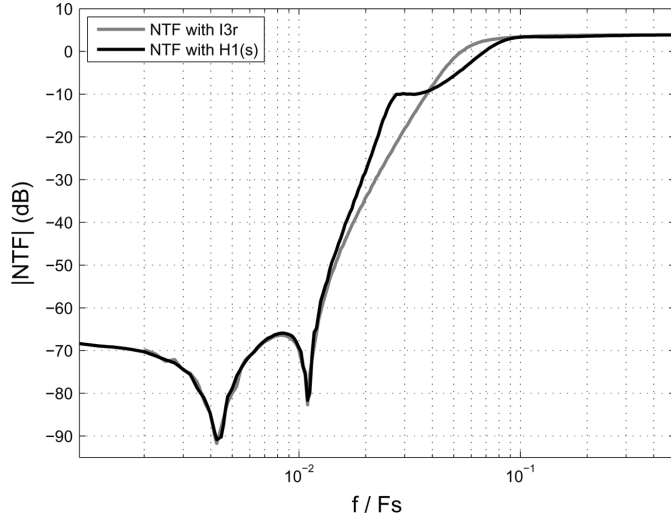


Fig. 8. Realized NTF of the compensated modulator, compared with a maximally flat NTF.

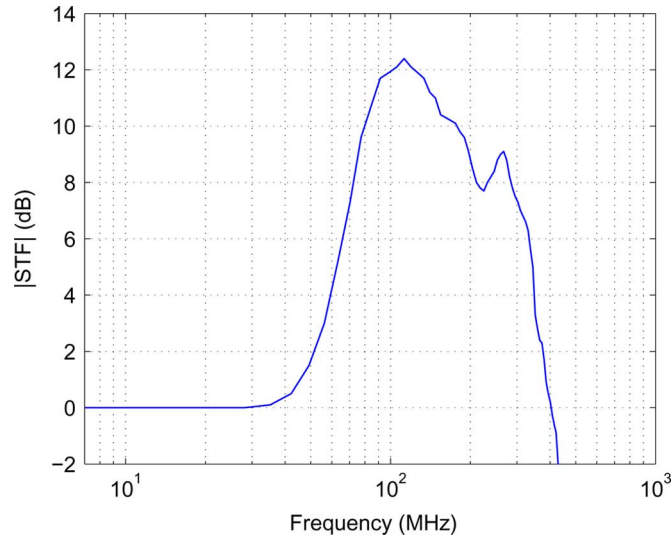


Fig. 9. Magnitude of the signal transfer function.

$H_1(s)$ in the compensation path is responsible for the kinks in the realized NTF. This is not of concern, however, as the in-band quantization noise and maximum stable amplitude are virtually the same as that achieved with a flat NTF. The magnitude of the STF (simulated) is shown in Fig. 9. It peaks at high frequencies, due to the extensive use of input signal feedforward (which results in small integrating capacitors). This peaking, which can be problematic when out-of-band inputs are present, has to be addressed by appropriate filtering prior to the modulator.

Other methods to stabilize the loop when an FIR-DAC is used have been proposed. The implementation in [11] uses a reduced gain for the first integrator to ensure stability, at the expense of an increased in-band quantization noise. Adopting this for our design was not an option, as the sampling frequency would have to be increased to achieve the desired SQNR. Another technique to compensate for the delay of the main FIR DAC involves using compensating FIR-DAC feed-ins into all integrators [10]. Unfortunately, the jitter noise injected by the compensation FIR-DACs begins to limit performance. In our CTDSM,

we avoid this by observing that the “useful” part of the compensation path has a bandpass spectrum. We exploit this by making the low frequency gain of the compensation path small, thereby reducing the in-band components of the signals injected by the compensation path. Further, the compensation path feeds into the third integrator (I_2), causing its nonidealities to be attenuated by the gain of two integrators when referred to the input.

IV. CIRCUIT DESIGN

A. Loop Filter

The single-ended equivalent of the modulator is shown in Fig. 10. Active-RC integrators are used due to their high output swings, reduced sensitivity to parasitics, and improved linearity. The high speed path around the quantizer is through I_1 (and, to a lesser extent, I_2). Therefore the parasitic poles of these integrators need to be at very high frequencies, necessitating power-hungry opamps. Power reduction is achieved in our design by implementing I_1 and I_2 using the assisted-opamp technique [15], where a replica of the integrator input current is injected into the output of the opamp. This reduces the opamp’s burden and enhances speed for a given power. In our design, current steering DACs—DAC_{a0} and DAC_{a2} are used to assist integrators I_1 and I_2 respectively. Assistance is only used for the “fast input currents” of I_1 and I_2 .

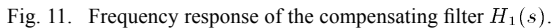
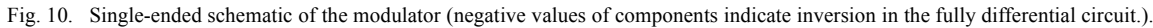
The FIR filter for the main DAC, which feeds into I_4 , has a transfer function denoted by $F(z)$. Although shown in Fig. 10 as an FIR filter driving a DAC, the actual implementation merges the filter and DAC into a semi-digital FIR structure, as shown in the lower part of the figure [13]. As discussed in the Introduction, resistor mismatch effects the frequency response of the filter, but does not impact linearity. The modulator is compensated for the delay introduced by the FIR DAC by the compensating path shown in blue. It consists of an FIR DAC $E(z)$ driving a filter $H_1(s)$, which is realized by I_5 . At low frequencies, the gain is proportional to $(R_{h1} + R_{h2})$. As frequency increases, C_{h3} approximates a short circuit, thereby causing R_{h1}, R_{h2}, R_{h3} to appear as a T-network, resulting in an increased gain. At very high frequencies, C_5 causes the response to fall again. In our design, the response peaks at about 100 MHz and asymptotically rolls off as $1/s$, as shown in Fig. 11. The output of $H_1(s)$ feeds into I_2 through an RC-series branch which acts like a high-pass filter. This shapes the jitter noise injected by the compensation path out of the signal band.

The modulator is scaled so that the first-order path saturates last. Simulations (and measurements) show that the modulator recovers after being subject to a signal beyond the maximum stable amplitude (MSA). As a safety feature (to prevent the loop from getting “stuck” in an oscillatory state), switches were placed across all capacitors so as to reset state variables if instability is detected.

Variations in integrator time constants modify the NTF and can potentially result in instability. To counteract this, the resistors and integrating capacitors are realized as digitally adjustable banks, and are manually controlled.

B. Operational Amplifiers

The opamps are two stage feedforward-compensated designs. Such opamps are inherently more power efficient than a Miller-



The macromodel of the first opamp is shown in Fig. 12(a). The high dc gain is obtained from a cascade of two transconductors g_{m1} and g_{m2} . An ac-coupled structure, realized using $r_b - c_b$ and g_{mff} , provides the high-speed feedforward path that stabilizes the opamp when it is enclosed in a negative feedback loop. A simplified schematic of the opamp is shown in

Thanks to the use of an FIR feedback DAC, the linearity requirements of the first integrator (I_4) are greatly relaxed. In our design I_4 consumes only 3 mA, while I_2, I_3 and I_5 consume 0.5 mA each. The opamp in I_1 needs to be fast, but has a small feedback factor. Hence, its current needed to be increased to 1 mA.

As discussed in the previous section, the use of an FIR feedback DAC significantly relaxes the design of the comparator, but operation at 3.6 GS/s in this technology is challenging. Fig. 13 shows the simplified circuit diagram and salient waveforms of the comparator used in this work. When the clock is low, the comparator is in the sampling phase, and the amplified input is held on the parasitic capacitances at the drains of M1 and M2. The common-mode voltage of the output nodes at the end of the sampling phase is close to the supply. During the

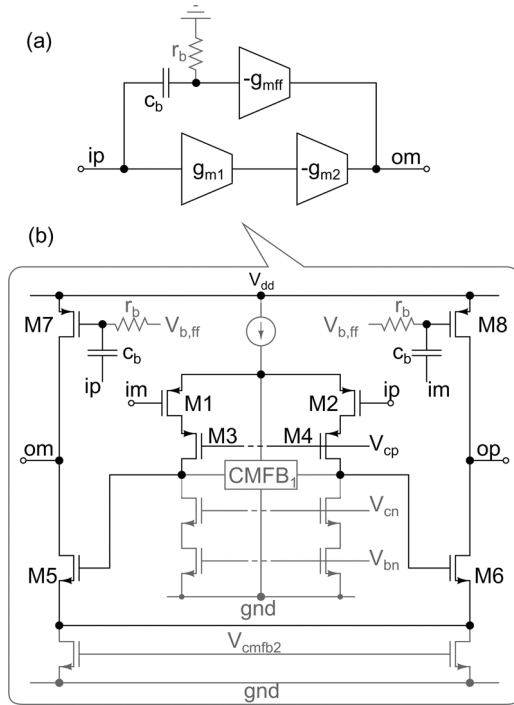


Fig. 12. (a) Macromodel and (b) schematic of the two-stage feedforward-compensated opamp used in the integrators.

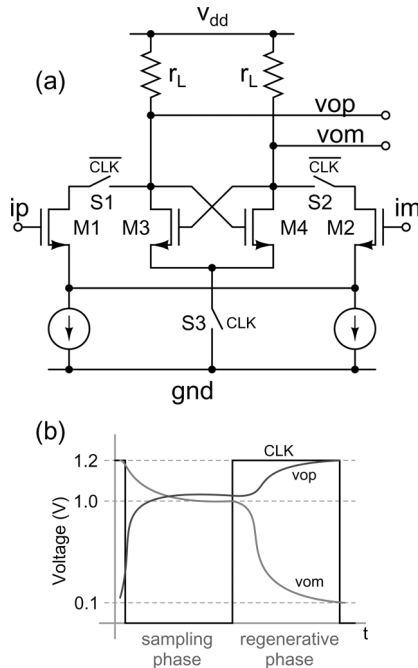


Fig. 13. (a) Simplified schematic of the comparator. (b) salient waveforms.

regeneration phase, M3 and M4 are turned on by connecting their sources to ground. This way, the current during the initial part of the latch phase is very high, thereby reducing the regenerative time constant. After regeneration, the current drawn from the supply is limited by the resistors. The large initial current in the regenerative pair enables lower power operation (for the same speed) when compared to a conventional CML latch. To demonstrate this, a CML latch with the same average

power dissipation was designed. Fig. 14 shows the regeneration currents in the proposed and CML latches for large and small differential inputs. The initial current is high in the proposed design and is of significance especially during metastable operation when the differential input voltage is very small. Fig. 15 shows the output voltages in the two cases for the same (small) input. The CML latch has a regenerative gain of less than one, while the proposed latch achieves a gain of about 4 at 3.6 GS/s.

The comparator uses a single phase clock, avoiding the need to generate non-overlapping clocks at such high speeds. This reduces the power consumption in the clock generation circuitry. Since the output of the comparator swings almost rail to rail (see Fig. 13(b)), it can directly drive the CMOS latch that senses its output. This is an added advantage with respect to CML latches, which need additional level-shifting stages (which dissipate power and add delay) to be able to drive CMOS gates.

D. Feedback DACs

The feedback DACs can be implemented in several ways. Current steering techniques are commonly used in high speed applications. However, these are particularly noisy in low voltage CMOS due to the limited headroom available for the transistors comprising the current source. Moreover, the parasitic capacitance at the source coupled node causes distortion. Hence, the current steering designs were avoided in the feedback DACs.

Switched resistors were used for all the feedback DACs due to their low noise operation. As shown in Fig. 16, such designs can be implemented with switches on the reference or virtual ground side. Both techniques seem identical at first sight—however, the distributed capacitance of the resistor and the parasitic capacitances at the virtual ground node need to be factored into the choice of the design.

Switching at the reference nodes causes delay in the DAC current waveforms due to the distributed-RC nature of the resistor. This is particularly problematic for large resistor values. However, an advantage of placing the switches at the references is that the loop gain around the opamp does not change with time (assuming instantaneous switching), and therefore does not degrade the modulator's linearity. It is thus seen that "reference side switching" is an apt choice where the DAC resistors are small, and/or when linearity is critical. Simulations in our technology indicate that resistors smaller than $10\text{ k}\Omega$ can be implemented with reference side switches. Accordingly, DAC₀, DAC₂ and DAC₄ are implemented in this fashion.

Placing the switches at the virtual ground nodes of the opamps avoids the need to charge and discharge the distributed parasitic capacitance of the resistors, and thereby results in high speed operation. However, the in-band noise and linearity of the modulator are degraded due to the switching action of the parasitic capacitance at the virtual grounds. Thus, such DACs are apt when speed is of primary concern (DAC₁ and DAC₃).

E. DAC Nonlinearity

The main feedback FIR-DAC (DAC₄) in our design consists of parallel resistors feeding currents into the virtual ground of the input opamp. A set of resistors, corresponding to a single tap

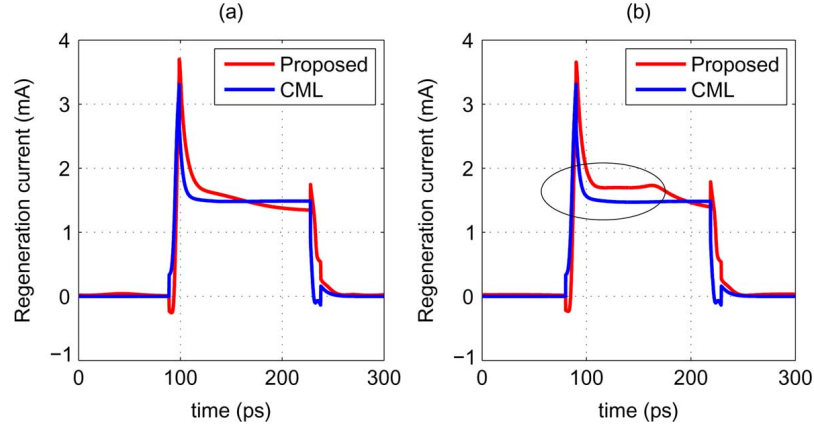


Fig. 14. Regeneration current of the proposed comparator and a CML comparator for (a) large inputs and (b) small inputs that result in metastability.

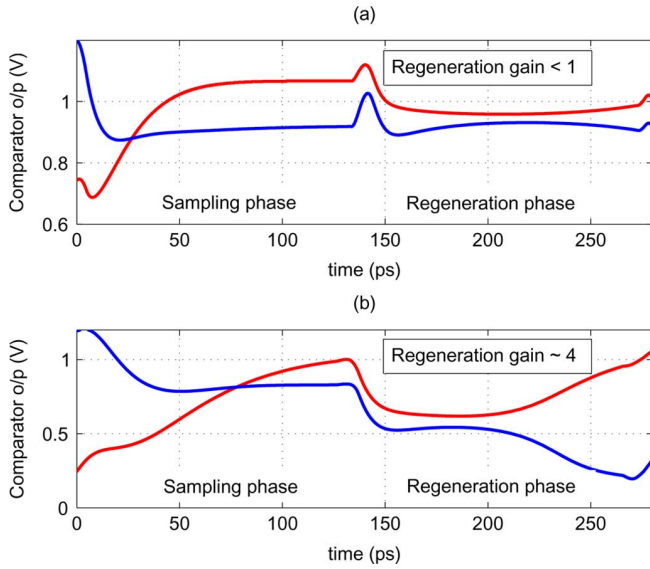


Fig. 15. Output waveforms for the same small inputs in (a) CML comparator and (b) the proposed design. Both comparators are clocked at 3.6 GHz.

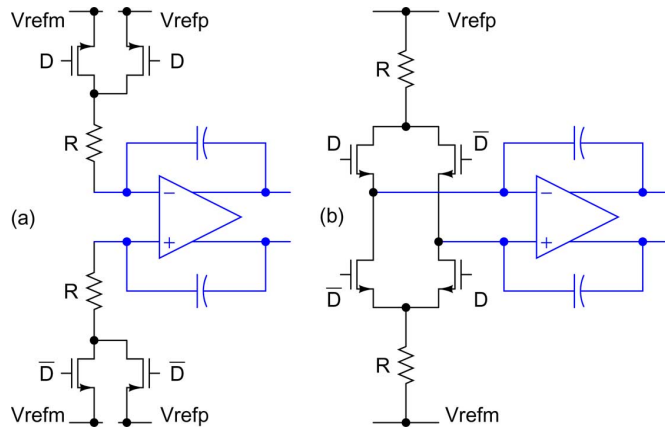


Fig. 16. Resistive DAC with switching on (a) the reference side and (b) the virtual ground side.

of the DAC, is shown in Fig. 17(a), where the DAC drive voltages are denoted by $vdac_p$ and $vdac_m$. The opamp is assumed to be ideal. At high speeds such as these, the rise and fall times

of $vdac_{p,m}$, are a significant fraction of the bit period, and are unequal. This results in asymmetric single-ended DAC current waveforms, as shown in Fig. 17(b). i_m (or i_p) can be thought of as the sum of an ideal ISI free wave form $i_{m,ideal}$ and an error waveform $i_{m,error}$. If $i_{m,error}$ was a linear function of the input data sequence ($v[k]$, taking on values -1 or 1), the error waveform at the rising and falling edges of v should be inverted versions of one another. However, it is apparent that this is not the case with rise/fall time asymmetry, and $i_{m,error}(t)$ can be expressed as follows:

$$i_{m,error}(t) = \sum_k \underbrace{(v[k] - v[k-1])p_r(t - kT)}_{\text{linear}} + \underbrace{|v[k] - v[k-1]|p_e(t - kT)}_{\text{nonlinear}} \quad (2)$$

where

$$p_e(t) = p_r(t) - p_f(t) \quad (3)$$

and $p_r(t)$ and $p_f(t)$ are the deviations from the ideal DAC waveform at the rising and falling edges, respectively, as shown in the figure. From the equations above, it is seen that the single-ended error current waveform has a nonlinear component dependent on $|v[k] - v[k-1]|$ and the difference between the rise and fall transitions. Due to the nature of the nonlinear term, we should expect second order distortion and an increase in the in-band noise floor. This is confirmed by the low-frequency PSDs shown in Fig. 17(d). However, for the differential DAC current without resistor mismatch ($r = 0$), the rise-fall transitions are identical, resulting in distortion-free operation, as shown in Fig. 17(c). Since linearity of the differential implementation is achieved by subtraction, one should expect some uncanceled second-harmonic and noise floor increase to manifest in a practical implementation.

Another aspect relevant in a single-bit modulator is the following. In a 2-level design, the transition density varies with signal level, being high for small modulator inputs, and decreasing as the input approaches full scale. From (2), we see that distortion is related to the number of transitions in the modulator output sequence. This means that the error added due to

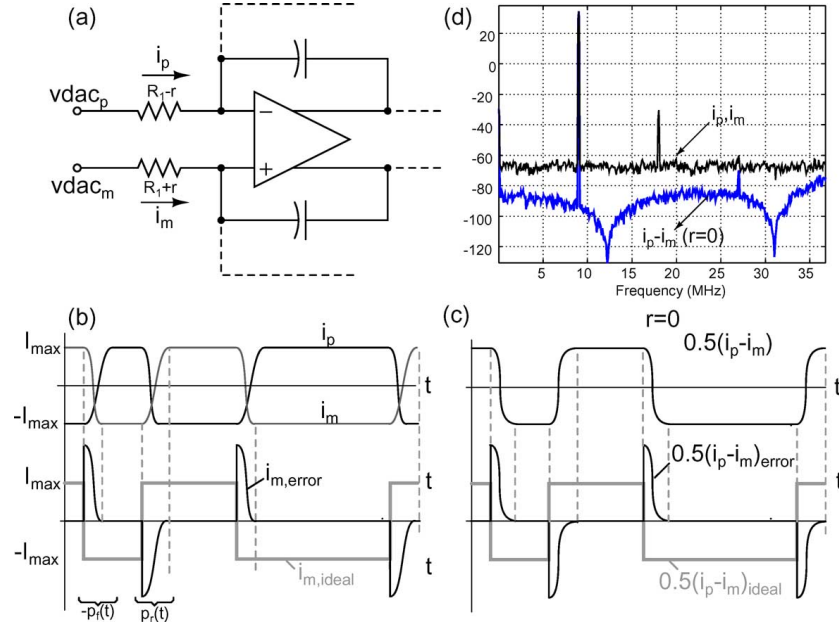


Fig. 17. (a) Resistive DAC feeding into the virtual shorts of an opamp (b) Current waveforms in single-ended arms (c) Differential current (d) Low frequency spectra of single ended and differential currents.

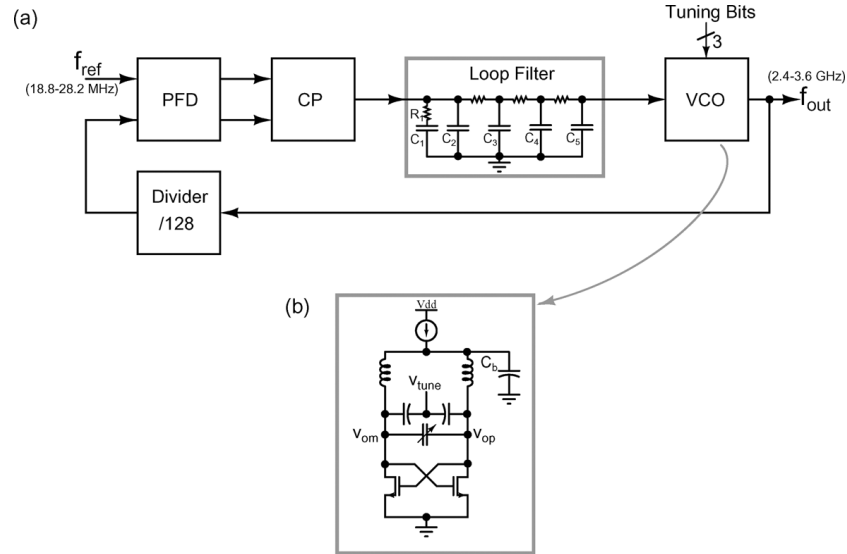


Fig. 18. (a) Block diagram of the LC-PLL and (b) schematic of the LC-VCO.

rise-fall asymmetry is amplitude dependent, causing second harmonic distortion, and an increased inband noise floor. This effect, which was also observed in our design, is shown in our measurement results (Section 5).

F. Phase Locked Loop (PLL)

An on chip LC-PLL was designed to provide a low jitter clock to the modulator. The PLL uses a low frequency external clock as the reference. A block diagram of the PLL is shown in Fig. 18(a). It consists of an LC VCO, a divide-by-128 circuit, Phase Frequency Detector (PFD), charge pump and loop filter. The differential LC VCO based on a cross coupled NMOS pair as shown in Fig. 18(c), is designed to meet the low phase noise requirement of the clock. The LC tank consists of two 2 nH

inductors, a MOS accumulation varactor and a digitally controllable MIM-capacitor bank. The inductors are circular and are wound in the thickest top most metallization layer of the process. The NMOS pair is biased with a current of 10 mA. A large MOS capacitor C_b bypasses the noise current from the PMOS current source. The center frequency of the VCO can be varied from 2.4-3.6 GHz. The sensitivity of the VCO is about 200 MHz/V.

The divide-by-128 circuit consists of seven divide-by-two cells in cascade. The PFD and charge pump are conventional designs. The series network $C_1 - R_1$ forms the usual integrating capacitor and stabilizing resistor. C_1 , which sets the PLL bandwidth, is about 1 nF and is external to the chip. In lock, the narrow output pulses of the charge pump cause spurs in the

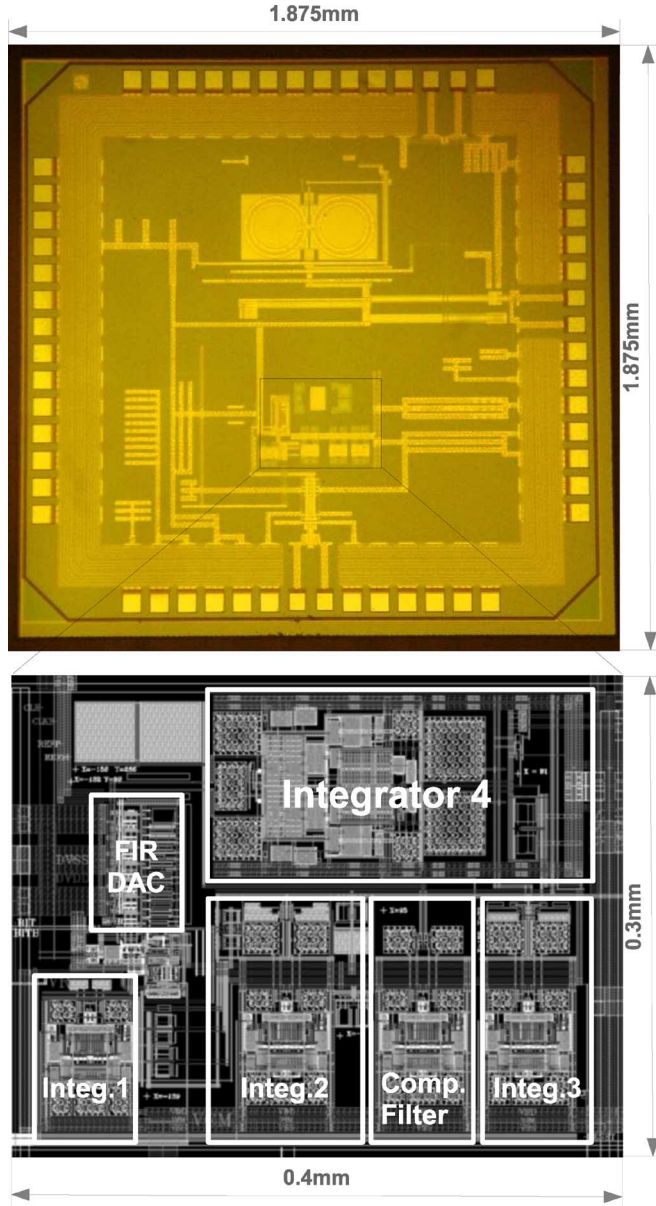


Fig. 19. Die photograph and modulator layout. The modulator occupies an active area of 0.12 mm^2 .

VCO output spectrum, which demodulate out-of-band quantization noise in the CTDSM into the signal band. To attenuate this spur, the control voltage is filtered by additional R-C networks. The loop bandwidth is set to about 100 kHz. The simulated phase noise of the VCO output is about -120 dBc/Hz at 1-MHz offset and less than -160 dBc/Hz at 100-MHz offset from the carrier.

V. MEASUREMENT RESULTS

The design was fabricated in a 90-nm CMOS process through Europractice and assembled in a 64-pin QFN package. The die photograph and layout snapshot are shown in Fig. 19. The modulator occupies 0.12 mm^2 and consumes 15 mW (12.5 mA) from a 1.2-V supply. The currents drawn by the analog, digital and reference circuitry are 5.5, 5.1, and 1.9 mA respectively. The PLL occupies 0.25 mm^2 and dissipates 14.4 mW.

A four-layer board was used to characterize the chip. The test setup is shown in Fig. 20. A passive band-pass filter (Allen Avionics BPS010P0B) was used to eliminate distortion from the input signal source. The sinusoid was then converted into a differential signal using a balun (Minicircuits-ADT1-1WT). The modulator output, which was brought out of the chip at full rate using an LVDS interface, was captured using a real-time oscilloscope (Agilent Infinium DSO80204B) and post processed to determine the output sequence. A 32-K Blackman–Harris window was used for spectral estimation. The CTDSM can be clocked by the on-chip PLL or an external source. Unless otherwise mentioned, all measurements reported below use the on-chip clock generator.

Fig. 21 shows the SNR/SNDR as a function of signal amplitude for a 10-MHz input. The peak SNR, SNR, and dynamic range in a 36-MHz bandwidth are 76.4, 70.9, and 83 dB, respectively. Fig. 22 shows the power spectral density (PSD) of the modulator output for a -4 dBFS tone at 10 MHz. We see that the SNDR is limited by the second harmonic.

A. Jitter Measurements

To observe the effect of using an FIR feedback DAC on jitter, the CTDSM was clocked with an FM modulated clock, generated by modulating a Minicircuits VCO with a sinusoidal source, as shown in Fig. 25. The rms jitter and the jitter frequency are determined by the amplitude and frequency of sinusoidal source. The frequency of the modulating signal was swept from 100 MHz to 1.8 GHz in steps of 100 MHz, while the rms jitter was fixed at 10% of the sampling period. The modulators input was zero for this measurement. Modulating the clock at a frequency f_m results in sidebands at these frequencies, which in turn result in a translation of the shaped quantization noise around f_m into the signal band. Fig. 26 shows the measured in-band noise as a function of f_m . Notice the minima in the noise around 600 and 800 MHz. These frequencies correspond to the zeros of the FIR filter driving the main feedback DAC, thereby confirming the “jitter-filtering” action of the FIR feedback technique.

Fig. 27 shows the maximum high-frequency clock jitter tolerated by the modulator to achieve an in-band noise-level of -80 dBFS . The figure compares the jitter tolerance of this design with the simulated tolerance of a CTDSM using a four-bit loop quantizer running at 1.2 GHz. The tolerance of a single-bit modulator without an FIR DAC is also shown for reference. It is seen that using an FIR DAC results in a jitter tolerance comparable with that of a four bit modulator operating at a third of the sampling rate and can handle ten times higher jitter than a single-bit design without an FIR DAC.

B. Measurement Summary and Performance Comparison

Table I summarizes the performance of our design. In a 36-MHz signal bandwidth, the design achieves a figure of merit (FOM) of 72.7 fJ/1v and the Schreier FOM of 176.8 dB, respectively. For a 25-MHz signal bandwidth, the modulator achieves 86-dB dynamic range, 80.2-dB peak SNR, and 73.3-dB peak SNDR, resulting in an FOM of 79.4 fJ/lev and Schreier FOM of 178.2 dB. Table II compares it with some state-of-the-art designs. Thanks to the architectural and circuit techniques

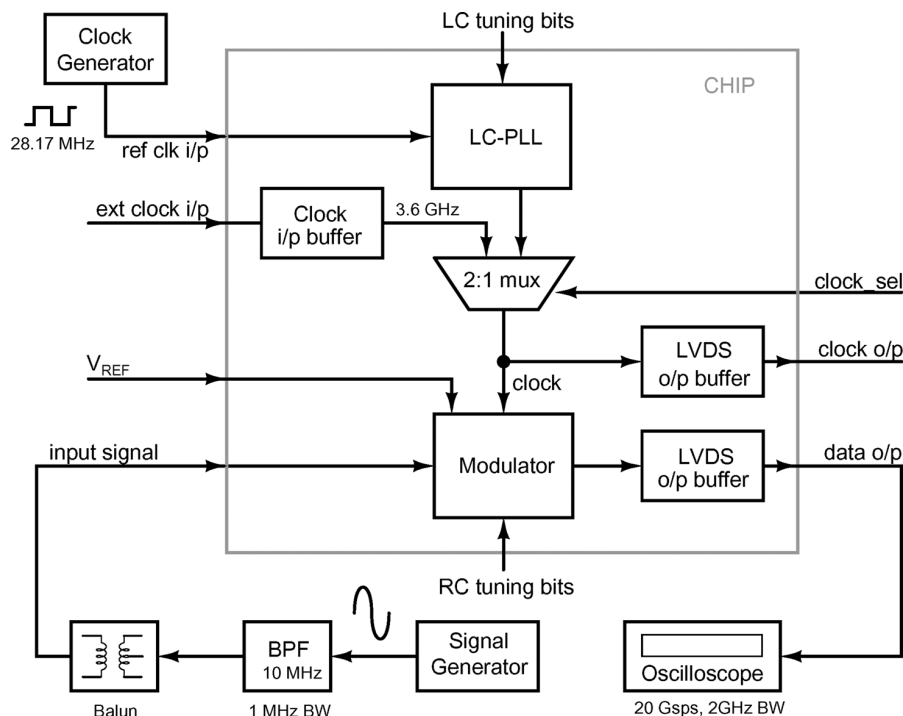


Fig. 20. Block diagram of the test setup.

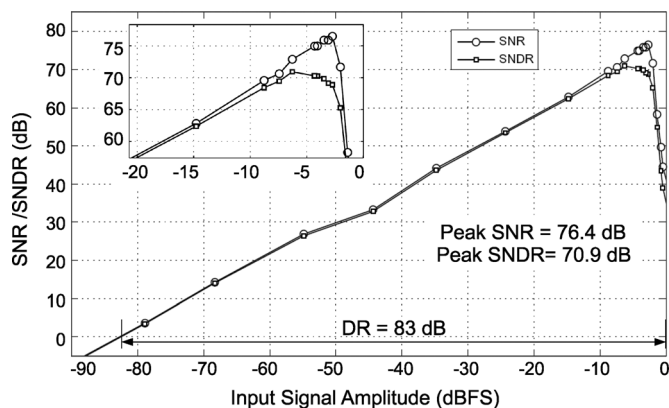
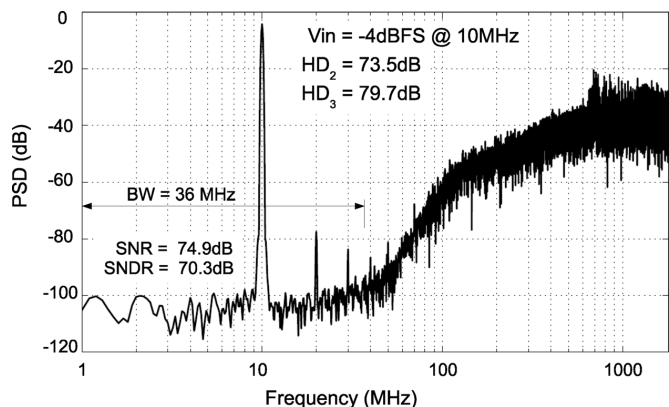


Fig. 21. SN(D)R in a 36-MHz bandwidth as a function of input amplitude for a 10-MHz input tone.

Fig. 22. PSD of the modulator output for a -4 -dBFS input signal.

employed in this design, we see that excellent power efficiency is achieved over a wide bandwidth. Fig. 23 shows the plot of bandwidth versus dynamic range for all the works reported at

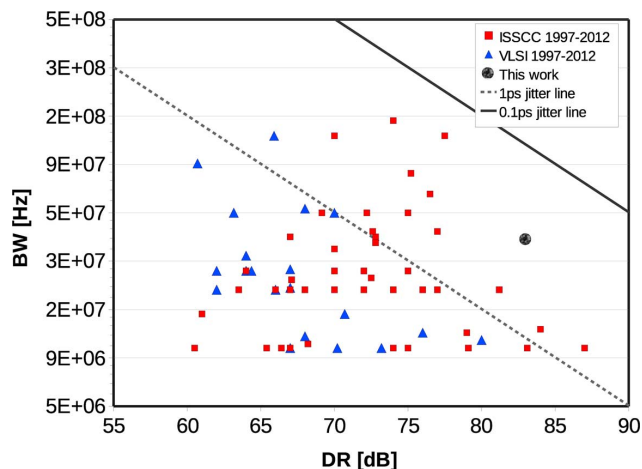


Fig. 23. Bandwidth versus DR comparison of this work with the works published in ISSCC and IEEE Symposium on VLSI Circuits in March 2012. Only designs with $DR > 60$ dB and $BW > 10$ MHz are shown.

the ISSCC and IEEE Symposium on VLSI Circuits upto March 2012 [16], which have $DR > 60$ dB and $BW > 10$ MHz. As seen from the plot, this work has highest signal bandwidth for converters that achieve $DR > 80$ dB. Fig. 24 is a scatter plot of ADC power (normalized to the Nyquist rate) versus SNDR for the converters in Fig. 23. We see that among ADCs that achieve $DR > 70$ dB, this work has the best FOM.

C. Correction for ISI Effects

A careful look at Fig. 21 reveals a kink in the SNDR-versus-amplitude curve around -55 dBFS. For inputs larger than this level, it was observed that the SNDR degraded due to an increase in the in-band noise floor. When the input amplitude approached about -8 dBFS, significant second harmonic distort-

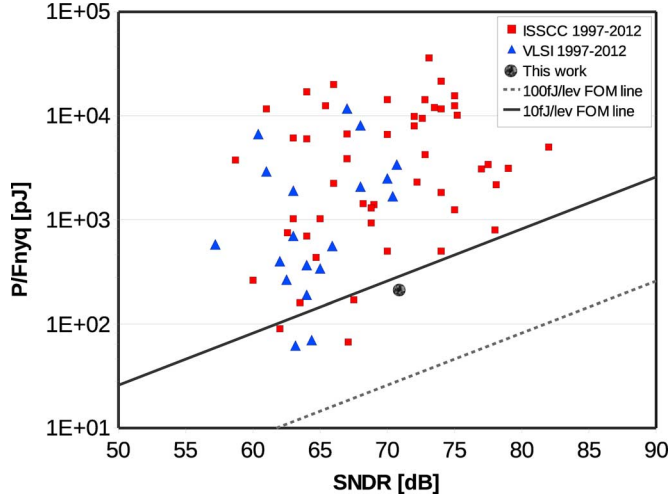


Fig. 24. Power (normalized to the Nyquist Rate) versus SNDR comparison of this work with those published at ISSCC and IEEE Symposium on VLSI Circuits up to March 2012. Only designs with DR > 60 dB and BW > 10 MHz are shown.

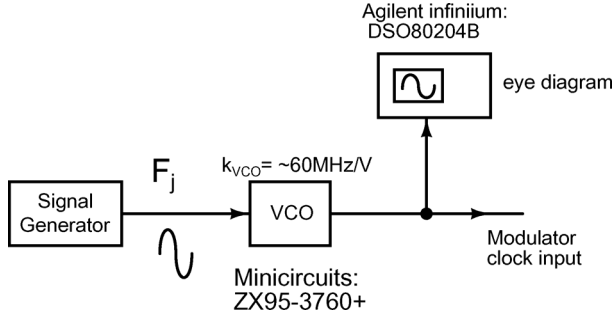


Fig. 25. FM modulated clock generation using an external VCO.

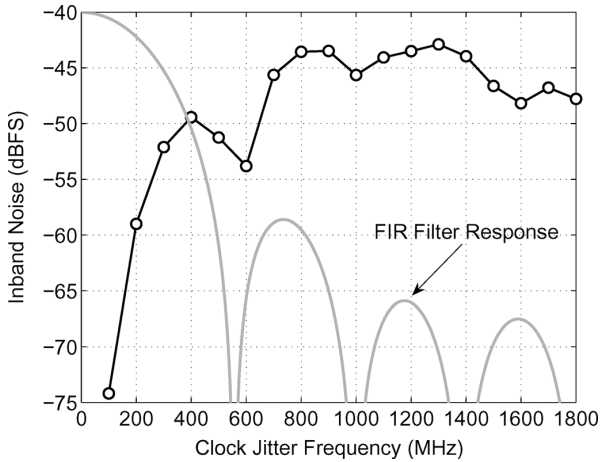


Fig. 26. Measure in-band noise of the modulator as the modulating frequency of the FM clock is varied. The rms jitter amplitude is maintained at 10% of the clock period.

tion was noticed, which degraded the SNDR further. As discussed earlier in this paper, this is most likely due to a combination of single-bit operation and nonlinearity induced by ISI in the feedback DAC. ISI causes distortion due to a difference in the rise and fall transitions of the differential feedback DAC waveform and can be corrected in several ways. At the circuit

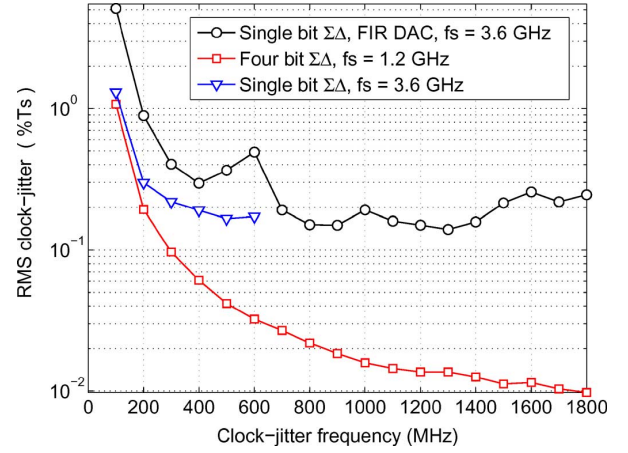


Fig. 27. Comparison of the jitter tolerance of this design (to achieve -80 dBFS idle channel inband noise) with the simulated tolerances of a four bit CTDSM operating at 1.2 GHz and a single-bit design without an FIR DAC.

TABLE I
PERFORMANCE SUMMARY

Technology	9-metal 90nm CMOS	
Active Area	0.12 mm ²	
Supply Voltage	1.2 V	
Sampling Rate	3.6 GHz	
Power Consumption	15 mW	
Bandwidth	25 MHz	36 MHz
Dynamic Range	86 dB	83 dB
Peak SNR	80.2 dB	76.4 dB
Peak SNDR	73.3 dB	70.9 dB
FOM _{DR}	18.4 fJ/1v1	18.0 fJ/1v1
FOM _{SNR}	35.9 fJ/1v1	38.6 fJ/1v1
FOM _{SNDR}	79.4 fJ/1v1	72.7 fJ/1v1
FOM _{Schreier}	178.2 dB	176.8 dB

$$FOM_{xx} = \frac{\text{Power}}{2 \cdot BW \cdot 2^{(xx-1.76)/6.02}}$$

$$FOM_{Schreier} = DR + 10 \log \left(\frac{BW}{\text{Power}} \right)$$

level, the dual RZ principle [17] can be used, though it could be challenge to implement at these speeds. A *post facto* technique, that can be implemented in the digital domain, is described using Fig. 28. Fig. 28(a) shows the current waveform of an NRZ DAC. The output spectrum is nonlinear due to ISI (which has been grossly exaggerated by making the fall time zero). An “ideal” current waveform which would be devoid of distortion is shown in Fig. 28(b), and the error between the ideal and actual wave forms is shown in part (c). We therefore see that the a single-bit CTDSM can be modelled as shown in Fig. 28(d), where the feedback DAC has a parallel nonlinear path which injects a pulse $h(t)$ at every rising edge. Ideally $h(t)$ should be zero—but in practice, $h(t)$ is a pulse with a small area, which causes an increase in the noise floor and adds second harmonic distortion. This can be corrected by adding a sequence which is α at the rising edge of the raw modulator output stream, as shown in Fig. 28(d). In our case, this was done in software with the idea of confirming the mechanism of distortion generation. α was chosen to be that value which minimized second harmonic

TABLE II
PERFORMANCE COMPARISON OF OUR WORK WITH SOME STATE-OF-THE-ART DESIGNS

	This Work	[1]	[2]	[3]	[4]	[5]	[6]
CMOS Node	90 nm	130 nm	90 nm	65 nm	90 nm	130 nm	45 nm
Area (mm ²)	0.12	1.2	0.15	0.15	1	0.45	0.9
Supply Voltage (V)	1.2	1.2	1.2	1.2/1.3	1.2	1.5	1.1/1.8
Sampling Rate	3.6 GHz	640 MHz	500 MHz	250 MHz	420 MHz	640 MHz	4 GHz
Power (mW)	15	20	8.0	10.5	27.9	87	256
Bandwidth (MHz)	36	20	25	20	20	20	125
DR (dB)	83	80	70	68	72	80	70
SNR _{max} (dB)	76.4	76	64	62	72	81.2	65.5
SNDR _{max} (dB)	70.9	74	63.5	60	70	78.1	65
FOM _{DR} (fJ/1vl)	18	61	62	128	214	266	394
FOM _{SNR} (fJ/1vl)	38.6	97	124	255	214	231	665
FOM _{SNDR} (fJ/1vl)	72.7	122	131	321	270	331	705
FOM _{Schreier} (dB)	176.8	170	164.9	160.8	160.6	163.6	156.9

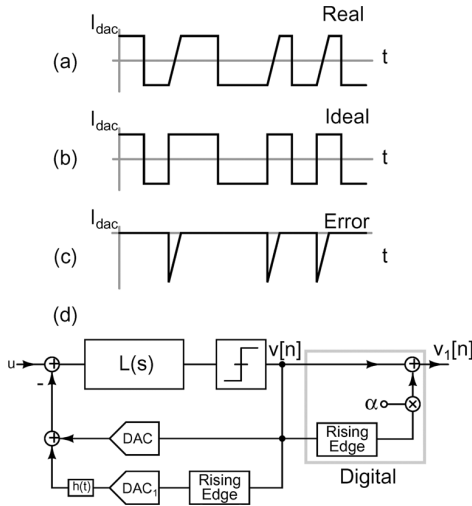


Fig. 28. (a) Waveform of an NRZ DAC with rise/fall time mismatch. (b) Ideal waveform. (c) Error. (d) Digital sequence that must be added to correct the error.

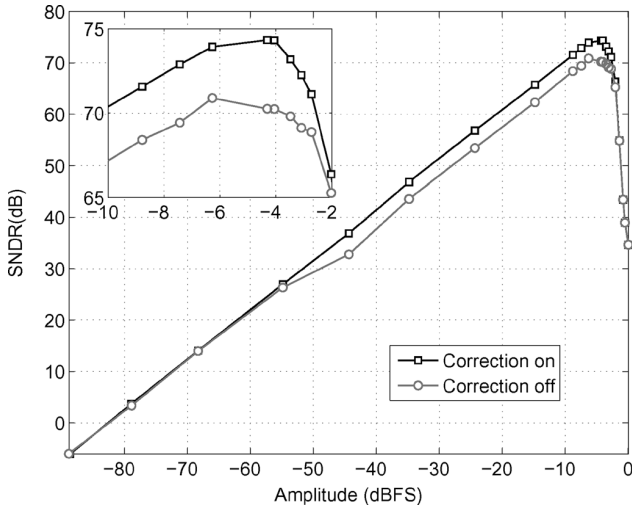


Fig. 29. Measured SNDR with and without applying digital correction for ISI.

distortion for a -4 -dBFS input amplitude at 10 MHz. Thereafter, α was fixed at this value, and the correction was applied

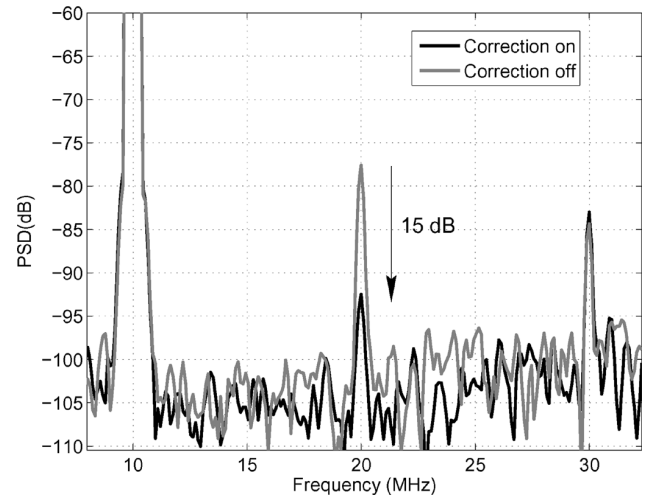


Fig. 30. PSD with and without correction. The second harmonic has significantly decreased, and the noise floor is lower.

to the data sequences corresponding to all the other amplitudes. The SNDR with and without correction are shown in Fig. 29. The peak SNDR is now 74.3 dB, and the kink in the SNDR curve is eliminated. The PSD for a -4 -dB input tone is shown in Fig. 30. We see that with correction, the second harmonic is smaller by 15 dB, and the noise floor is lower. The peak SNR with correction is 77.5 dB. Similar improvements were seen when correction was applied *with the same* α when the input frequency was changed from 10 to 5 MHz. These results confirm the origin of the distortion components, and indicate that power efficiency of the design can be further improved.

VI. CONCLUSION

We demonstrated that single-bit modulators with FIR feed-back DACs can achieve wide bandwidths in a power efficient manner. The FIR DAC technique combines the power efficiency and inherent linearity of a single-bit quantizer with the low jitter sensitivity of multibit operation. Compensating the modulator loop for the delay introduced by the FIR filter is challenging and

was addressed using a mostly analog compensation path. In addition, we described high speed circuit techniques that complement the advantages of the proposed architecture. A testchip incorporating these design principles achieved a dynamic range of 83 dB and a peak SNDR of 70.9 dB in 36-MHz bandwidth while sampling at 3.6 GS/s in 90-nm CMOS. Consuming 15 mW from a 1.2-V supply, the converter has an FOM of 72.7 fJ/lvl, thereby making a compelling case for further work in this direction.

ACKNOWLEDGMENT

The authors acknowledge the Department of Science and Technology, Government of India, for funding and Euro-practice for chip fabrication. They would also like to thank Dr. N. Krishnapura, Indian Institute of Technology, Madras, for useful discussions.

REFERENCES

- [1] G. Mitteregger, C. Ebner, S. Mechnig, T. Blon, C. Holuigue, E. Romani, A. Melodia, and V. Melini, "A 14 b 20 mW 640 MHz CMOS CT $\Delta\Sigma$ ADC with 20 MHz signal bandwidth and 12 b ENOB," in *Int. Solid State Circuits Conf. Dig. Tech. Papers*, Feb. 2006, pp. 131–140.
- [2] J. Kauffman, P. Witte, J. Becker, and M. Ortmanns, "An 8.5 mW continuous-time $\Delta\Sigma$ modulator with 25 MHz bandwidth using digital background DAC linearization to achieve 63.5 dB SNDR and 81 dB SFDR," *IEEE J. Solid-State Circuits*, vol. 46, no. 12, pp. 2869–2881, Dec. 2011.
- [3] V. Dhanasekaran, M. Gambhir, M. Elsayed, E. Sanchez-Sinencio, J. Silva-Martinez, C. Mishra, L. Chen, and E. Pankratz, "A 20 MHz BW 68 dB DR CT sigma delta ADC based on a multi-bit time-domain quantizer and feedback element," in *Int. Solid State Circuits Conf. Dig. Tech. Papers*, Feb. 2009, pp. 174–175.
- [4] P. Malla, H. Lakdawala, K. Korngay, and K. Soumyanath, "A 28 mW spectrum-sensing reconfigurable 20 MHz 72 dB SNR 70 dB SNDR DT sigma delta ADC for 802.11 n/WiMAX receivers," in *Int. Solid State Circuits Conf. Dig. Tech. Papers*, Feb. 2008, pp. 496–497.
- [5] M. Park and M. Perrott, "A 0.13 μm CMOS 78 dB SNDR 87 mW 20 MHz BW CT sigma delta ADC with VCO-based integrator and quantizer," in *Int. Solid State Circuits Conf. Dig. Tech. Papers*, Feb. 2009, pp. 170–171.
- [6] M. Bolatkale, L. Breems, R. Rutten, and K. Makinwa, "A 4 GHz continuous-time $\Delta\Sigma$ ADC with 70 dB DR and 74 dBFS THD in 125 MHz BW," *IEEE J. Solid-State Circuits*, vol. 46, no. 12, pp. 2857–2868, Dec. 2011.
- [7] J. Cherry and W. Snelgrove, "Clock jitter and quantizer metastability in continuous-time delta-sigma modulators," *IEEE Trans. Circuits Syst. II, Analog Digit. Signal Process.*, vol. 46, no. 6, pp. 661–676, Jun. 1999.
- [8] S. Pavan, "Alias rejection of continuous-time $\Delta\Sigma$ modulators with switched-capacitor feedback DACs," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 58, no. 2, pp. 233–243, Feb. 2011.
- [9] W. Lee, "A novel higher order interpolative modulator topology for high resolution A/D converters," Master's thesis, Dept. Electr. Eng., Mass. Inst. of Technol., Cambridge, MA, 1987.
- [10] O. Oliaei, "Sigma-delta modulator with spectrally shaped feedback," *IEEE Trans. Circuits Syst. II, Analog Digit. Signal Process.*, vol. 50, no. 9, pp. 518–530, Sep. 2003.
- [11] B. Putter, " $\Sigma\Delta$ ADC with finite impulse response feedback DAC," in *Int. Solid State Circuits Conf. Dig. Tech. Papers*, 2004, pp. 76–77.
- [12] P. Shettigar and S. Pavan, "A 15 mW 3.6 GS/s CT-delta sigma ADC with 36 MHz bandwidth and 83 dB dynamic range in 90 nm CMOS," in *Int. Solid State Circuits Conf. Dig. Tech. Papers*, Feb. 2008, pp. 498–501.
- [13] D. Su and B. Wooley, "A CMOS oversampling D/A converter with a current-mode semidigital reconstruction filter," *IEEE J. Solid-State Circuits*, vol. 28, no. 12, pp. 1224–1233, Dec. 1993.
- [14] S. Pavan, "Efficient simulation of weak loop filter nonlinearities in continuous-time oversampling converters," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 57, no. 8, pp. 233–243, Aug. 2010.
- [15] S. Pavan and P. Sankar, "Power reduction in continuous-time delta-sigma modulators using the assisted opamp technique," *IEEE J. Solid-State Circuits*, vol. 45, no. 7, pp. 1365–1379, Jul. 2010.
- [16] B. Murmann, "ADC Performance Survey 1997–2012," 2012.
- [17] R. Adams and K. Nguyen, "A 113 dB SNR oversampling DAC with segmented noise-shaped scrambling," *IEEE J. Solid-State Circuits*, vol. 33, no. 12, pp. 1871–1878, Dec. 1998.



Pradeep Shettigar (M'12) received the B.E. degree in electronics and communication engineering from the Sri Jayachamarajendra College of Engineering, Mysore, in 2006. He is currently working toward the M.S. degree in electrical engineering at the Indian Institute of Technology, Madras, India.

His research interests are in the area of high-speed data conversion and signal processing.



Shanthi Pavan (SM'12) received the B.Tech. degree in electronics and communication engineering from the Indian Institute of Technology, Madras, India, in 1995, and the M.S. and Sc.D. degrees from Columbia University, New York, in 1997 and 1999, respectively.

From 1997 to 2000, he was with Texas Instruments, Warren, NJ, where he worked on high-speed analog filters and data converters. From 2000 to June 2002, he worked on microwave ICs for data communication with Bigbear Networks, Sunnyvale, CA. Since July 2002, he has been with the Indian Institute of Technology, Madras, India, where he is now a Professor of electrical engineering. His research interests are in the areas of high-speed analog circuit design and signal processing.

Dr. Pavan is a Fellow of the Indian National Academy of Engineering. He is the Deputy Editor-in-Chief of the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS: PART I—REGULAR PAPERS, and served on the editorial board of the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS PART II—EXPRESS BRIEFS from 2006 to 2007. He serves on the technical program committee of the International Solid State Circuits Conference. He is the recipient of the Shanti Swarup Bhatnagar Award (2012), the IEEE Circuits and Systems Society Darlington Best Paper Award (2009), the Swarnajayanti Fellowship (from the Government of India), the Young Faculty Recognition Award from IIT Madras (for excellence in teaching), the Technomenter Award from the India Semiconductor Association and the Young Engineer Award from the Indian National Academy of Engineering (2006).